Method	Pretraining	Classification (macro-F1)	Detection (mAP)
Chance level	–	16.67	7.84
Classical acoustic features			
Spectral features	–	33.52 ± 2.06	26.34 ± 1.17
MFCCs	–	44.20 ± 2.20	33.91 ± 1.77
Mean spectrogram	–	54.95 ± 2.15	47.80 ± 2.10
Off-the-shelf pretrained models			
AVES-core	General audio (AudioSet, FSD50K)	67.60 ± 2.11	57.39 ± 2.07
BioLingual	Audio-text (AnimalSpeak)	70.98 ± 2.12	66.52 ± 2.23
AVES-bio	Animal vocalizations (AudioSet, VGGSound)	75.13 ± 1.96	65.01 ± 2.29
Pretrained on OpenWhistle (ours)			
AVES-bio‡	OpenWhistle (from scratch)	79.89 ± 1.81	76.17 ± 1.97
Wav2Vec2.0	OpenWhistle (from scratch)	$\mathbf{81.40 \pm 1.78}$	$\mathbf{79.59 \pm 1.83}$
Continual pretraining on OpenWhistle (ours)			
AVES-bio†	AVES-bio checkpoint $\rightarrow$ OpenWhistle	75.24 ± 2.01	73.87 ± 2.06

Table 1: Performance on whistle-type classification (balanced, macro-F1) and whistle-type detection (mAP). Models are grouped by representation and pretraining strategy. All use linear probing with a logistic regression classifier on frozen representations. Scores are reported as percentages. Each entry gives the full-test metric ± the standard deviation over 1,000 bootstrap resamples of the test set. †Continual pretraining at 16 kHz with multi-band encoding; ‡trained natively at 44.1 kHz.